

CS495 Software Project Proposal: Digital Watershed

Joel Robinson & Yohann Gouin

2026-01-20

1 Client Information

By sharing this client information and the rest of this document, you are stating that this client has provided this project as something they want (not something you created and asked if they wanted), and that they are interested in having you complete this project for your capstone.

1. Client name: Billy Friebele
2. Client title: Program Director of Studio Art
3. Client email address: wefriebele@loyola.edu
4. Client employer: Loyola University Maryland
5. How you know the client: Mr. Friebele was a previous professor that had taught me in Studio Arts.

2 Project Description

2.1 Overview

This project aims to design and create a digital system that organizes, preserves, and attempts to visualize a multi-modal body of research from the client. As the research expands, pulling in documents, maps, audio, video, 3D scans, and other materials, the variety and volume of media can become difficult to manage. The problem is the lack of a unified, and intuitive tool that can collect, tag, structure, and navigate these research threads. The purpose of the project is to create a sustainable web-based application that functions as a digital watershed, allowing the user to store items from research streams and organize them into larger conceptual categories. This container will support the clients ongoing artistic practice by providing an archive for inspiration, and also offer an interactive representation showing how individual pieces of research flow into broader themes. The client wants this project because it provides a long term stability for research workflow, a structured method for managing creative archives, and a dynamic way to display the connections alongside the artwork inspired from it. It aims to understand, explore, and communicate the relationships within the clients research.

2.2 Key Features

- Centralized Digital Archive
- Flexible Tagging & Categorization
- Item Search & Filtering
- Visualization of Tagged Relationships
- Sustainable Web Application Architecture for longevity.

2.3 Why this Project is Interesting

This project is interesting because its a mix between computer science, art, and research, making it a unique capstone that transforms a creative research process into a digital ecosystem of data for connections to happen with. The challenge of integrating multiple media types and being able to tag and scrub data from them to form relationships, and generating interactive visualizations allows for in depth work across several CS areas. The creative space between technology and the studio arts is something that is very compelling, and the client's idea of "revealing what cant be seen" through their art is something that connects with the relational tagging and visualization the product wants to do.

2.4 Areas of CS required

The subfields of computer science that are most likely to be relevant to this would be web development, because there would be frontend and backend engineering for managing data processing, authentication, API services, for the web app to function. There is also database systems, as it must store, index, categorize, and retrieve any and all media formats it holds efficiently. There is also information visualization, connecting media within the containers to visualize relationships between data, for future artistic purposes.

2.5 Potential Concerns and Questions

The aspect of the project that has me the most unsure, but is equally the most interesting, is the data visualization when the app begins to form connections via tags with the data, and the direction it should go. Is something like mind-maps of visualization too easy and should something deeper be chosen? What is the potential flourishes that could be added to make it non-generic? This is something we have never interacted with before and would be entirely new to us.

2.6 Comparison to Draft

This project is different from the projects before, as neither of us were able to get in contact with good clients before the proposal was due, and this Watershed Web App was the one option we had. This client is new, because the clients we had fielded in assignments prior were unresponsive when reaching out, which led to us looking more internally in Loyola for past professors or staff that would have needs that we could accommodate for our capstone project.

3 Requirements

3.1 Non-Functional Requirements

NFR1	Color Scheme	Visual / Design	The color scheme for the app should be white and blue.
NFR2	Password Hashing	Security	All passwords shall be hashed and salted using bcrypt.
NFR3	Error Logging	Maintainability	The system shall log errors with sufficient context to allow developers to debug or resolve issues locally.
NFR4	Recycle Bin	Reliability	The system shall support soft-deletion so items can be recovered if needed.
NFR5	Data Visualization Performance	Performance	The dashboard shall load and display the total number of items in under 2 seconds.
NFR6	Data Copying	Reliability	The system shall maintain a protected data copy to restore from in the event of data corruption.
NFR7	Item Search Responsiveness	Reliability	Search results shall be shown progressively instead of waiting for the entire search to complete.
NFR8	Item Placeholder Logo	Visual / Design	Items without a preview shall be displayed with a standard placeholder logo.
NFR9	Item Backup	Reliability	The system shall perform consistent backups to ensure data preservation in case of errors.

Table 1: Non-Functional requirements

3.2 Functional Requirements (User Stories)

ID	Story Title	Points	Description
S1	Search Media	2	As a user, I want to search for specific media, so that I can find exactly what I need.
S2	Manage Tags	8	As a user, I want to create, read, update, and delete tags, so that I can appropriately maintain and tag research.
S3	Manage Items	8	As a user, I want to create, read, update, and delete items, so that I can maintain my data archive consistently.
S4	Media Preview	8	As a user, I want to view items with the correct preview player, so that I can consume the data within my app.
S5	Create Projects	8	As a user, I want to create projects within the web app, so that I can categorize my research items by project.
S6	Monitor Storage	2	As an admin, I want to monitor storage usage and media count, so that I can understand capacity across projects.
S7	Audit Log	5	As an admin, I want an audit log of critical actions, so that I can trace changes over time.
S8	Manage Users	8	As an admin, I want to manage users, so that the system remains secure and maintainable.
S9	Shared Links	5	As a user, I want to control visibility with shared links, so that I can share parts of my data container and projects.
S10	Recently Viewed	2	As a user, I want a recently viewed list, so that I can return to recent items or research threads.
S11	Login	2	As a user, I want the capability to log in, so that I can access my account.
S12	Logout	2	As a user, I want the capability to log out, so that I can securely leave my account.
S13	External Links	2	As a user, I want to attach external links to items, so that I can reference sources without uploading files.
S14	Tag Relationships	8	As a user, I want to define relationships between tags, so that items can be related beyond basic tagging.
S15	Filter Items	5	As a user, I want to filter items by tag, category, or media type, so that I can narrow results for specific goals.
S16	Sort Results	3	As a user, I want to sort results by criteria such as newest or oldest, so that I can browse efficiently.
S17	Batch Upload	5	As a user, I want to batch upload files, so that I can ingest large collections efficiently.
S18	Timeline View	5	As a user, I want to view my research project on a timeline, so that I can understand how my ideas evolved over time.
S19	Upload Media Types	3	As a user, I want to upload various media types, so that I can preserve my full research collection in one place.
S20	Visual Connections	5	As a user, I want to see visual connections between research items and themes, so that I can understand relationships.
S21	Interactive Visualization	5	As a user, I want to interact with visualizations, so that I can explore research streams dynamically.
S22	Link Artworks	3	As a user, I want to link finished artworks to source research items, so that I can show inspiration paths.
S23	Network Graph View	8	As a user, I want to view research items as nodes in a network graph, so that I can see all connections at once.
S24	Graph Interaction	5	As a user, I want to filter, search, and interact with the network graph, so that I can navigate complex relationships.
S25	Media Metadata	5	As a user, I want to add media-specific metadata, so that each item captures relevant contextual information.
S26	Batch Tagging	3	As a user, I want to apply tags to multiple items simultaneously, so that I can organize large batches efficiently.
S27	Export Data	5	As a user, I want to export selected items and metadata, so that I can share collections or create backups.

Table 2: Functional requirements as User Stories.

ID	Story Title	Points	Description
S28	Notes	2	As a user, I want to add personal notes and annotations, so that I can record insights during research.
S29	Duplicate Detection	5	As a user, I want the system to detect duplicate uploads, so that my archive remains clean.
S30	Custom Dashboard	5	As a user, I want to customize my dashboard with widgets, so that I can quickly access relevant research areas.

Table 3: Functional requirements as User Stories.

4 System Design

4.1 Architecture

[Which type of software architecture are you team following? Layered architecture, MVC, other? What are the main modules for your software?] Our system will use MVC architecture. The View will be the web interface where users browse projects, search and filter, upload media, and interact with the project visualizations. The Controller will be our backend that receives requests from the UI, does authentication, input validation, and actions like item creation, tags, exporting, and graphing. The Model will represent the applications data using MongoDB, including all the different items within it, responsible for data persistence and retrieval.

4.2 Diagrams

[CS482, on sprints/iterations 2-3, you need to create and update a diagram (check the assignment for which type of diagram). On CS496, since before sprint/iteration 1 you should have a class diagram and keep it up-to-date.]

4.3 Technology

The project will be implemented as a web app using JavaScript as the programming language. The frontend will be built with React, providing an interface for browsing research items, managing projects and tags, and interaction with the visualizations. Styling will be handled using Bootstrap CSS. Data Visualizations will use JS visualizations libraries capable of rendering graphs and timelines. The backend will use Node.js with express, following its MVC pattern to handle routing/validation/authentication. MongoDB with Mongoose as the object-document mapper for defining schema and structure will be used. Bcrypt will be used for hashing and salting of passwords. Jest will be used for testing, and SuperTest will work the API-level testing.

4.4 Coding Standards

The project follows consistent coding standards to ensure readability, maintainability, and reliability throughout. All code will follow a consistent naming convention, with JavaScript variables and functions written in camelCase, React components written in PascalCase, and MongoDB models named using singular nouns while collections remain plural. API routes will follow RESTful naming conventions, and environment-specific configuration will be managed using environment variables rather than hard-coded values.

4.5 Data

[What is the main structure of your data? In SQL-like databases, this would be the planning of the main tables, their attributes, and interactions with other tables (basically an ER diagram). In NoSQL databases, this would be the main collections and general attributes of the JSON you will store in each collection.]

4.6 UI Mocks

[Define and draw/sketch/code the main UIs your user will interact with in your software. Add your UI mocks here and a short caption about it. Do not forget about the main forms and CRUD UIs.]

5 Iterations

5.1 Iteration Planning

[In CS496, you plan all iterations beforehand. In CS482, you update the planning here at each iteration.]

Iteration	Dates	Stories	Points
1	1/29 - 2/10	S3 Manage Items, S4 Media Preview, S11 Login, S12 Logout, S13 External Links, S19 Upload Media Types, S25 Media Metadata, S28 Notes & Annotations	32
2	2/11 - 2/24	S1 Search Media, S2 Manage Tags, S5 Create Projects, S10 Recently Viewed, S15 Filter Items, S16 Sort Results, S17 Batch Upload, S26 Batch Tagging	36
3	2/25 - 3/17	S14 Tag Relationships, S18 Timeline View, S20 Visual Connections, S22 Link Artworks	21
4	3/18 - 3/31	S21 Interactive Visualization, S23 Network Graph View, S24 Graph Interaction	18
5	4/1 - 4/14	S6 Monitor Storage, S7 Audit Log, S8 Manage Users, S9 Shared Links, S27 Export Data, S29 Duplicate Detection, S30 Custom Dashboard	35
Total:			142

Table 4: Iteration Planning for Incremental Deliveries

5.2 Iteration/Sprint 1

5.2.1 Planning

This iteration plans to complete S3 Manage Items, S4 Media Preview, S11 Login, S12 Logout, S13 External Links, S19 Upload Media Types, S25 Media Metadata, and S28 Notes / Annotations, for a total of 32 points. The focus of this sprint is delivering a functional viable product that allows users to securely authenticate, upload and manage research items, and view their content with appropriate previews and metadata. Completing these stories creates the core functionality required for the system to be usable before higher-level organization and visualization features are introduced.

5.2.2 Work Done

[Which stories did you complete in this iteration/sprint. Which ones did you partially complete? Who worked on which story? You may elaborate in paragraph(s) to add more detail about the work done.]

5.2.3 Testing Coverage

[Testing is very important. Show your coverage here. Is this coverage good enough? Explain why you think so. Is it not good enough? Explain a plan to increase the coverage. You may also elaborate on why some artifacts do not undergo much testing. If the testing changed from the last iteration, explain the reasons.]

5.2.4 Retrospective & Reflection

[What were the pitfalls, challenges, and issues you had in this iteration? How can you address them to improve the process in the next iteration? Did anything not go according to plan? Why so and how to avoid the same mistake? Write a personal reflection on what you learned in this iteration (even if a small technical thing like Database storage).]

5.3 Iteration/Sprint 2

5.3.1 Planning

This iteration includes S1 Search Media, S2 Manage Tags, S5 Create Projects, S10 Recently Viewed, S15 Filter Items, S16 Sort Results, S17 Batch Upload, and S26 Batch Tagging, totaling 36 points. The goal of this sprint is to improve organization and efficiency as the volume of stored research grows. By adding search, filtering, sorting, tagging, and project grouping, users can quickly locate and manage large collections of items, while batch operations and recently viewed features streamline the research workflow.

5.3.2 Work Done

[Which stories did you complete in this iteration/sprint. Which ones did you partially complete? Who worked on which story? You may elaborate in paragraph(s) to add more detail about the work done.]

5.3.3 Testing Coverage

[Testing is very important. Show your coverage here. Is this coverage good enough? Explain why you think so. Is it not good enough? Explain a plan to increase the coverage. You may also elaborate on why some artifacts do not undergo much testing. If the testing changed from the last iteration, explain the reasons.]

5.3.4 Retrospective & Reflection

[What were the pitfalls, challenges, and issues you had in this iteration? How can you address them to improve the process in the next iteration? Did anything not go according to plan? Why so and how to avoid the same mistake? Write a personal reflection on what you learned in this iteration (even if a small technical thing like Database storage).]

5.4 Iteration/Sprint 3

5.4.1 Planning

This iteration plans S14 Tag Relationships, S18 Timeline View, S20 Visual Connections, and S22 Link Artworks, for a total of 21 points. The focus of this sprint is on helping users understand relationships and the evolution of their research over time. Timeline views, tag relationships, and visual connections provide context and structure, while linking finished artworks back to research items highlights the connection between inspiration and creative output.

5.4.2 Work Done

[Which stories did you complete in this iteration/sprint. Which ones did you partially complete? Who worked on which story? You may elaborate in paragraph(s) to add more detail about the work done.]

5.4.3 Testing Coverage

[Testing is very important. Show your coverage here. Is this coverage good enough? Explain why you think so. Is it not good enough? Explain a plan to increase the coverage. You may also elaborate on why some artifacts do not undergo much testing. If the testing changed from the last iteration, explain the reasons.]

5.4.4 Retroespective & Reflection

[What were the pitfalls, challenges, and issues you had in this iteration? How can you address them to improve the process in the next iteration? Did anything not go according to plan? Why so and how to avoid the same mistake? Write a personal reflection on what you learned in this iteration (even if a small technical thing like Database storage).]

5.5 Iteration/Sprint 4

5.5.1 Planning

This iteration includes S21 Interactive Visualization, S23 Network Graph View, and S24 Graph Interaction, totaling 18 points. The primary objective of this sprint is to deliver advanced interactive visualizations that allow users to explore complex relationships within their research. Representing items as nodes in a network graph and enabling interaction through zooming, filtering, and navigation brings the project's digital watershed concept fully to life.

5.5.2 Work Done

[Which stories did you complete in this iteration/sprint. Which ones did you partially complete? Who worked on which story? You may elaborate in paragraph(s) to add more detail about the work done.]

5.5.3 Testing Coverage

[Testing is very important. Show your coverage here. Is this coverage good enough? Explain why you think so. Is it not good enough? Explain a plan to increase the coverage. You may also elaborate on why some artifacts do not undergo much testing. If the testing changed from the last iteration, explain the reasons.]

5.5.4 Retroespective & Reflection

[What were the pitfalls, challenges, and issues you had in this iteration? How can you address them to improve the process in the next iteration? Did anything not go according to plan? Why so and how to avoid the same mistake? Write a personal reflection on what you learned in this iteration (even if a small technical thing like Database storage).]

5.6 Iteration/Sprint 5

5.6.1 Planning

This iteration plans S6 Monitor Storage, S7 Audit Log, S8 Manage Users, S9 Shared Links, S27 Export Data, S29 Duplicate Detection, and S30 Custom Dashboard, for a total of 35 points. The focus of the final sprint is system refinement, administrative oversight, and long-term usability. These features support sharing, exporting, data integrity, customization, and security, ensuring the system is maintainable and scalable.

5.6.2 Work Done

[Which stories did you complete in this iteration/sprint. Which ones did you partially complete? Who worked on which story? You may elaborate in paragraph(s) to add more detail about the work done.]

5.6.3 Testing Coverage

[Testing is very important. Show your coverage here. Is this coverage good enough? Explain why you think so. Is it not good enough? Explain a plan to increase the coverage. You may also elaborate on why some artifacts do not undergo much testing. If the testing changed from the last iteration, explain the reasons.]

5.6.4 Retroespective & Reflection

[What were the pitfalls, challenges, and issues you had in this iteration? How can you address them to improve the process in the next iteration? Did anything not go according to plan? Why so and how to avoid the same mistake? Write a personal reflection on what you learned in this iteration (even if a small technical thing like Database storage).]

6 Final Remarks

6.1 Overall Progress

[Have you completed everything? If so, present evidence on how you brought value to your client, and the overall client satisfaction. Otherwise, estimate how much progress you done and how long it would take to finish this project.]

6.2 Project Reflection

[Your personal reflection on the project. What lessons did you learned. What would you have done differently. How can you do better work in future projects? You may write this as a team or per person (or both)]

Appendix

[Appendix section if needed]